

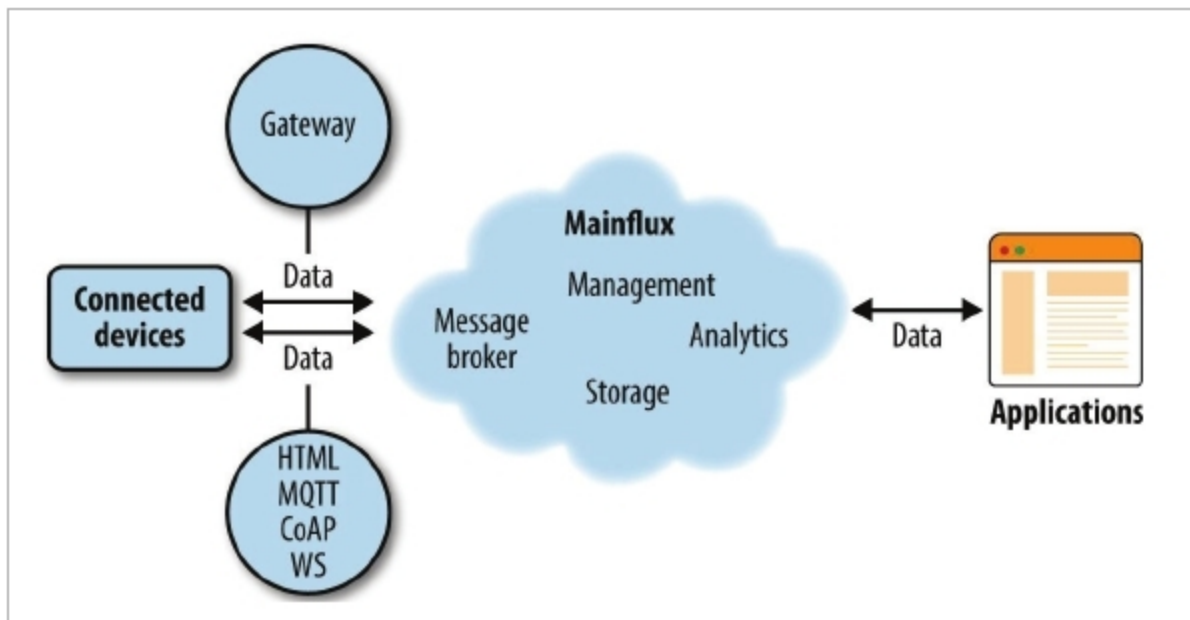


Figure 1: Location of Mainflux inside the IoT system

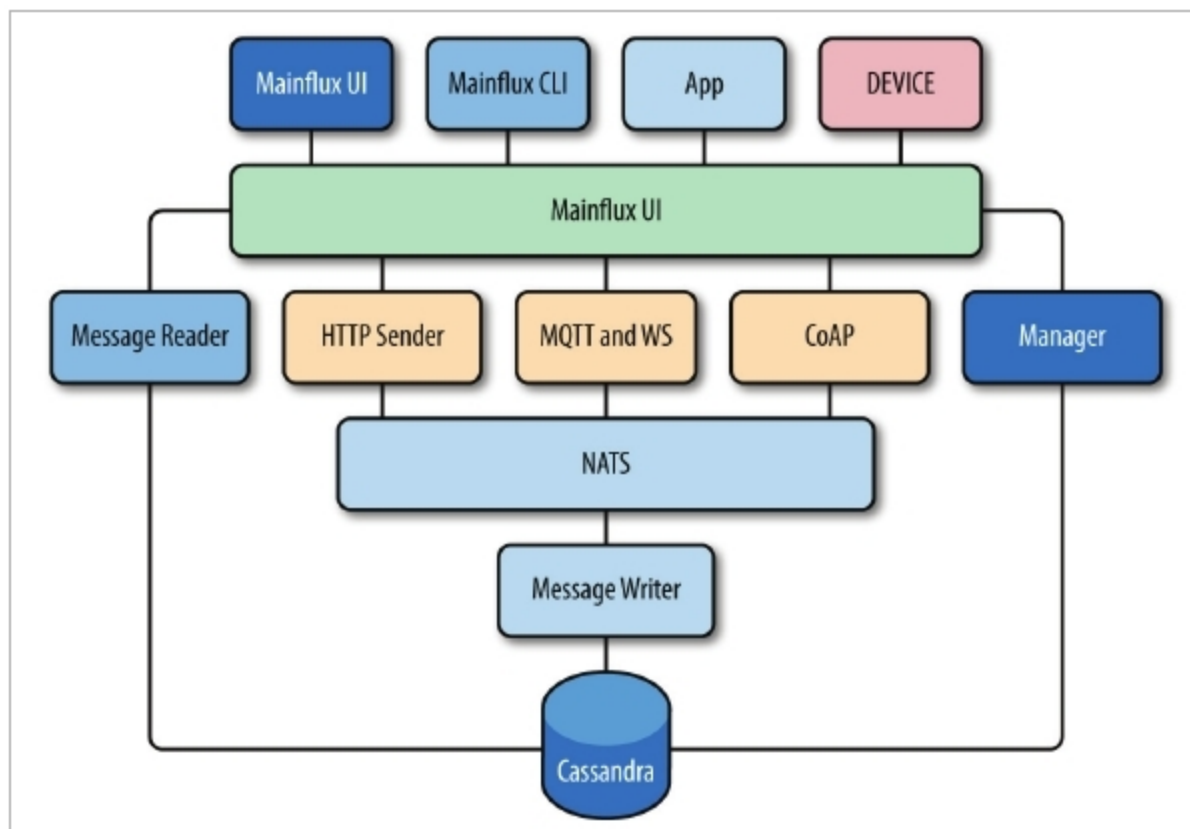


Figure 2: The key constituent elements of Mainflux

- Application manager (App)—accepts application connections on northbound interface
- User manager (Mainflux UI, Mainflux CLI)—provides user management for the applications
- Time-series storage engine (Message Reader, Message Writer, Cassandra)—stores and queries measurement data points in the time-series format
- Complex Event Processing (uses the system event bus NATS)—consumes the incoming time-series streams, and can automate triggers and actions based on a configurable set of rules.

Apart from messaging, Mainflux contains three main domains: user, thing and channel.

User represents the real user of the system. A user can access the platform through the access token, which can be obtained via an email id and password. Once logged, users can access or manage the resources (things and channels) by CRUD operations, and can define access control policies by connecting them.

Thing represents the applications or devices connected to Mainflux, which use the platform for message exchange with another device.

Channel represents the way through which a message is sent; simply put, we can say it is a communication channel.

Installation

Before installing Mainflux, install the prerequisites given below:

- Docker
- Docker Compose

Once the above components are installed successfully, the next step is to download the source code. You can do this in two ways — with or without the Git tool. I prefer to do it with the Git tool as it will download all the source code in the appropriate folder. But if you do this without the Git tool, you will have to download the Zip file from the GitHub repository, and extract the source code to the destination directory on your machine. Once it is done, you are ready for the next step.

We will now use the terminal to navigate to the folder where we have downloaded or extracted the Mainflux source code. For example, for my machine I used the `/home/user/` directory. If the Git tool is used in the previous step, then there should be a `/home/user/mainflux` directory as your path. Else, if you have used the Zip file and extracted it to your machine, then a folder might be created with the name as `mainflux-master`. Navigate to that source code directory. The next step is to start the Mainflux core services by executing the command:

```
docker-compose -f docker/docker-  
compose.yml up -d
```

Assuming that the above command is executed successfully, the Mainflux IoT platform should be up and running.

Some points to note

- Docker composition is defined under your project `root/docker/docker-compose.yml` file. You can customise this file according to your specific needs.